SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 1.5: Respiration — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.5: Respiration
Driving Question	DRIVING QUESTION: How do living organisms — from the yeast in our uji to the cells in our own muscles — release energy from food, and how can we use these invisible processes to create useful products and solve real problems in our communities?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **The Sealed Thermos of Uji That Keeps Changing — Predict & Anchor**	Students observed a sealed thermos of uji (fermented maize porridge) that had been left overnight. They noted visible changes: a sour smell when opened, slight bubbling or fizzing, a change in taste, and a slight warmth compared to room temperature. These changes served as the anchoring phenomenon for the entire sub-strand.	Students learned that breathing (upumuaji) and cellular respiration (upumuaji wa seli) are two distinct processes. Breathing is the physical movement of air in and out of the lungs, while cellular respiration is a chemical process occurring inside every living cell that releases energy (ATP) from food (glucose). The changes in the uji — sourness, bubbles, warmth — are evidence that invisible chemical processes are happening inside the microorganisms in the porridge.	Teacher note: At this stage, students are not expected to know the full mechanism of respiration. The goal is to anchor curiosity. Accept all predictions without correction — record them on the Driving Question Board (DQB). Use Socratic questioning: 'What do you think caused the sour smell? Where did the bubbles come from? Did the uji use energy or release energy?' Emphasise that the thermos analogy (sealed system, no visible input) makes the invisible process of cellular respiration puzzling and worth investigating. This lesson sets up the need-to-know that will drive all subsequent lessons.
Lesson 2 **Aerobic Respiration: Releasing Energy from Ugali in Every Cell**	Students examined a diagram and short reading showing a Kenyan long-distance runner (e.g., Eliud Kipchoge) during a marathon. They traced the journey of glucose from ugali (carbohydrate source) through digestion, into the bloodstream, and into muscle cells — where oxygen from breathing is used to release energy. A simple limewater demonstration confirmed CO_2 is exhaled during breathing, linking	Students learned the overall word and symbol equation for aerobic respiration: Glucose + Oxygen → Carbon dioxide + Water + Energy (ATP). $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + ATP. They learned that aerobic respiration occurs in the mitochondria and releases a large amount of usable energy (approximately 38 ATP molecules per glucose). Oxygen is essential; without it, the cell must switch to	Teacher note: Stress the distinction between the equation as a summary and the actual multi-step process. Use the analogy of burning charcoal (mkaa) — the energy in charcoal is released when it reacts with oxygen, just as glucose 'burns' in a controlled way inside mitochondria. Cold-call at least 4 students to identify inputs and outputs. Ask: 'If Kipchoge's muscle cells run out of oxygen at km 30,

	breathing to cellular chemistry.	a less efficient pathway.	what do you predict happens to his performance?' This foreshadows anaerobic respiration in Lesson 3. Post the aerobic equation on the DQB as the first confirmed answer.
Lesson 3 **Caught in the Act: Investigating Anaerobic Respiration in Yeast**	Students conducted a hands-on experiment: yeast + sugar solution was sealed in a bottle with a balloon stretched over the neck. Over 15–20 minutes, they observed the balloon inflating (CO_2 production) and recorded data. A control (boiled/dead yeast) showed no balloon inflation. Limewater turning milky confirmed CO_2 was the gas produced. Students also noted a faint alcohol smell from the active flask.	Students learned that: (1) yeast carries out anaerobic respiration (fermentation) in the absence of oxygen; (2) the products are carbon dioxide (CO_2) and ethanol (alcohol); (3) the word and symbol equation is: Glucose → Ethanol + Carbon dioxide + (small amount of) Energy. $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 + ATP$. This is called alcoholic fermentation. The process releases far less ATP than aerobic respiration (~2 ATP vs ~38 ATP).	Teacher note: This lesson is the critical empirical anchor for anaerobic respiration. Ensure students connect the balloon inflation directly to the CO_2 in the uji thermos from Lesson 1 — 'We now have evidence for what caused the bubbles!' Emphasise the NGSS practice of Analysing and Interpreting Data: require students to compare the experimental and control flasks quantitatively (balloon diameter or qualitative scale). Ask: 'Why did the boiled yeast produce no gas? What variable did we control?' Allow 2 minutes wait time for written predictions before results are revealed. Record the alcoholic fermentation equation on the DQB.
Lesson 4 **Anaerobic Respiration in Animals & Plants: Products, Equations, and the Sour Truth in Our Uji**	Students compared two scenarios: (1) a Kenyan farmer's muscles during intense digging (jembe work) — feeling the burn after sustained effort; (2) uji that has been left to ferment overnight becoming sour. They read a short case study explaining that muscle cells produce lactic acid under oxygen debt, while yeast/plant cells produce ethanol and CO_2. A pH indicator strip showed the fermented uji was more acidic than fresh uji.	Students learned that anaerobic respiration in yeast and plants (alcoholic fermentation) converts glucose into ethanol and carbon dioxide, releasing a small amount of energy. In animals (including humans), anaerobic respiration produces lactic acid (not ethanol): Glucose → Lactic acid + (small amount of) Energy. $C_6H_{12}O_6 \rightarrow 2C_3H_6O_3 + ATP$. The accumulation of lactic acid causes the burning sensation in muscles and leads to 'oxygen debt' — the body must continue breathing rapidly after exercise to oxidise the lactic acid.	Teacher note: The sour taste of fermented uji is direct evidence of acid production — use this to make lactic acid tangible. Draw a comparison table on the board: Aerobic vs. Anaerobic (yeast) vs. Anaerobic (animals) — substrate, products, ATP yield, location. Cold-call 5 students to fill in one cell each. Ask: 'Why does the farmer keep panting after he stops digging, even though the digging is over?' (oxygen debt). Clarify the common misconception that lactic acid = pain permanently — emphasise it is recycled in the liver when oxygen becomes available again. Update the DQB to answer: 'Why is the uji sour?'
Lesson 5 **The ATP Story: Where Does the Energy Actually Go?**	Students watched or read a short illustrated account of ATP (adenosine triphosphate) as the cell's 'energy currency' — analogous to M-Pesa tokens that can be spent anywhere in the cell. They examined diagrams showing ATP being produced during respiration and 'spent' during muscle contraction, active transport, and biosynthesis. A worked example compared	Students learned that ATP is the universal energy carrier in all living cells. Aerobic respiration produces approximately 36–38 ATP per glucose molecule; anaerobic respiration produces only 2 ATP per glucose molecule. The difference explains why aerobic respiration is far more efficient for sustained activity. Energy released from ATP hydrolysis ($ATP \rightarrow ADP + P_i$ + energy)	Teacher note: The M-Pesa analogy is powerful for Kenyan students — just as M-Pesa converts money into a usable token, respiration converts glucose chemical energy into ATP 'tokens' the cell can spend. Avoid over-mathematising the ATP count at this level; focus on the qualitative contrast (38 vs. 2). Ask: 'If a matatu engine used anaerobic respiration instead of aerobic

	the ATP yield of aerobic vs. anaerobic respiration numerically.	powers all cellular processes — movement, growth, active transport, and synthesis of molecules like proteins.	combustion, how far do you think it would travel on the same amount of fuel?' (Elicit: much less far — only ~5% efficiency). Connect back to Kipchoge: 'Why must marathon runners breathe aerobically for most of the race?' Post the ATP yield comparison on the DQB.
Lesson 6 **Factors Affecting the Rate of Respiration: Why Does Uji Ferment Faster in Mombasa Than in Kericho?**	Students designed and observed a structured investigation comparing yeast activity (balloon inflation rate) at different temperatures: cold water (~10°C, simulating Kericho/Aberdare highlands), room temperature (~25°C, simulating Nairobi), and warm water (~37–40°C, simulating Mombasa coast). They recorded balloon diameter at 5-minute intervals and plotted a graph. A class discussion also addressed the effects of glucose concentration and pH.	Students learned that the rate of respiration in living organisms is controlled by at least four key factors: (1) Temperature — increasing temperature speeds up enzyme activity and respiration rate, up to an optimum (~37–40°C for human cells); beyond the optimum, enzymes denature and the rate drops sharply. (2) Glucose (substrate) concentration — more substrate increases the rate until enzymes are saturated. (3) Oxygen availability — more oxygen drives faster aerobic respiration; absence forces anaerobic respiration. (4) pH — enzymes function optimally at a specific pH; extremes denature them.	Teacher note: The Mombasa vs. Kericho framing is a powerful contextual hook — students from both regions will immediately relate to why food spoils faster at the coast. Emphasise that all four factors work through enzymes — this is the mechanistic link. Require students to draw a correctly labelled rate-vs-temperature graph showing the rise, optimum, and sharp fall after denaturation. Common misconception to address: 'Higher temperature is always better' — correct this explicitly using the denaturation concept. Ask: 'What would a tea farmer in Kericho need to do differently to ferment uji compared to a fisherman in Kisumu?' Cold-call 3 students. Update DQB: 'Why does uji ferment faster in hot weather?'
Lesson 7 **DQB Revision & Economic Importance of Anaerobic Respiration**	Students reviewed their Driving Question Board entries from Lessons 1–6 and assessed which questions had been answered and which remained open. They then examined case studies and images of Kenyan industries that depend on anaerobic respiration: (1) uji/togwa fermentation (home and commercial scale); (2) Tusker/EABL beer brewing in Nairobi; (3) Kenyan bread and mandazi baking (yeast-leavened); (4) biogas production from livestock waste in rural Rift Valley farms; (5) silage production for dairy cattle feed in Kericho.	Students learned that anaerobic respiration is not only a cellular survival mechanism — it is the biochemical engine behind major sectors of Kenya's agro-processing economy. Alcoholic fermentation by yeast is harnessed in brewing, baking (CO_2 causes dough to rise), and production of bioethanol fuel. Lactic acid fermentation by bacteria is used in yoghurt, fermented milk (mursik among the Kalenjin), and silage. Biogas (methane + CO_2) produced by anaerobic bacteria digesting organic waste provides clean cooking energy for rural Kenyan households.	Teacher note: This lesson bridges science and real-world application — a core CBE competency. Ask students to identify which of these industries exists in their own county or community. For biogas, show a simple diagram of a biogas digester and ask: 'What is the substrate? What microorganism is responsible? What is the useful product?' (Organic waste → methane → cooking fuel). Revisit the DQB formally: read each original question aloud, cold-call a student to answer, and mark it as 'answered' or 'still investigating.' Identify any remaining gaps that will be closed in Lesson 8. This metacognitive DQB review is a high-value pedagogical moment — allow 10 full minutes for it.
Lesson 8 **Model Building & Project Showcase: Final**	Students presented their cumulative explanatory models (diagrams, flowcharts, or annotated posters) built and revised	Students consolidated that: (1) both aerobic and anaerobic respiration release ATP from organic substrates (primarily glucose); (2)	Teacher note: The model showcase is a performance task — assess models using the three criteria: accuracy (correct

Explanation of Respiration	across all 8 lessons. Models were required to show: the inputs and outputs of aerobic and anaerobic respiration, the role of mitochondria, ATP as energy currency, environmental factors affecting rate, and at least one real-world application. A final class discussion revisited the original anchoring phenomenon — the sealed thermos of uji.	aerobic respiration is far more efficient (36–38 ATP) and requires oxygen, producing CO_2 and water; (3) anaerobic respiration produces 2 ATP and generates ethanol + CO_2 (yeast/plants) or lactic acid (animals); (4) the respiratory quotient (RQ = CO_2 produced ÷ O_2 consumed) can be used to identify the substrate being respired — RQ = 1 for carbohydrates, >1 for anaerobic, <1 for fats; (5) these processes underpin both biological function and economic production across Kenya.	equations and products), completeness (all lesson concepts represented), and application (real-world Kenyan connection). Require every model to explicitly answer the driving question. During the final class discussion, return physically to the thermos of uji from Lesson 1 and cold-call students to now give a full scientific explanation of every change observed — sourness (lactic/acetic acid), bubbles (CO_2 from yeast fermentation), warmth (energy/ATP released as heat). If time allows, pose an extension question: 'Could you design a small biogas digester using kitchen waste from your home? What would you need?' This connects to the NGSS practice of Designing Solutions and the CBE value of community problem-solving.